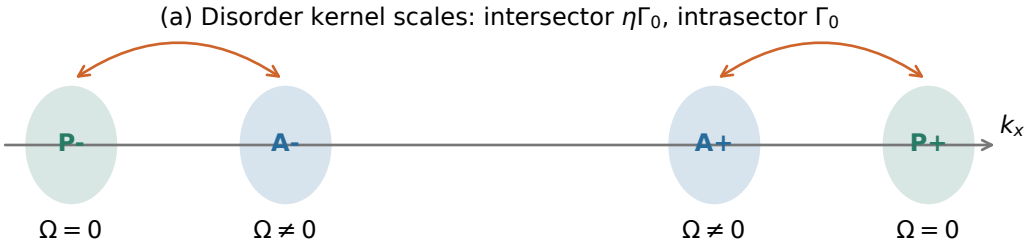

(b) Fixed order of added physical degrees of freedom

1 Active reference current	2 Active valley distortion	3 Passive current	4 Passive valley population
$g_{A0}/\|g_{A0}\|$	$(I - u_1 u_1^T)p_A$	$b_P/\|b_P\|$	$p_P/\|p_P\|$
Full uncoupled current response	Orthogonal valley change	Driven longitudinal motion	Opposite partner populations